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Research and implementation of multiscale dynamic terrain

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***Abstract*—In the real-time rendering phase of terrain, there is the possibility of dynamic deformation of terrain, such as dynamic of surface elevation and color texture. In order to better effect the surface appearance, we use higher resolution to reflect the terrain deformation. This results in multiscale dynamic deformation in the terrain. In view of this situation, the maximum change region, namely the deformation region, is determined after the accuracy is improved. Then, we establish our own binary tree and DAG structure in the deformation region. Furthermore, the nested sphere of error metric is used to avoid the generation of crack and T-connection. Finally, in order to give priority to the deformation region, the radius of nested sphere of error metric is used to constrain the error radius. The experiment proves that the algorithm can well reflect the deformation effect of the deformation area and meet the requirements of real-time terrain rendering after the local region has multiscale dynamic deformation.**

Keywords—multiscale dynamic deformation, LOD, deformation region

I. INTRODUCTION

The rendering of 3d terrain is one of the important research fields in virtual visualization[1, 2]. The rendering of 3d terrain mainly focuses on massive data storage, real-time rendering speed and realistic degree. In order to meet the requirements of real-time rendering of terrain, a lot of work was put into pre-processing. Therefore, the efficiency of the algorithm is improved, in order to achieve the goal of terrain rendering speed. But in the rendering of the terrain, the terrain of the original terrain changes in the render. Just as a car will leave a rut when it passes through soft ground, a bomb will leave a crater when it explodes on the ground. We said that, in the process of terrain rendering, the original terrain elevation values of surface properties such as color, texture of changing terrain called dynamic terrain. At present, there are more and more researches on dynamic terrain [3, 4].

Compared with static terrain visualization, dynamic terrain visualization introduces additional computing and storage overhead. ROAM algorithm is modified and extended[3] to meet real-time requirements and reduce storage overhead. However, the problem of grid discontinuity between small regions is not solved. Therefore, in the study of real-time rendering of dynamic terrain, the guarantee of real-time rendering speed and the rendering effect of real-time deformation of dynamic terrain are mainly discussed.

Used in the dynamic terrain rendering of terrain data, usually with 1 m to 30 m or more sampling points. However, in real life, the tracks on the surface of the feet are usually less than 0.3 meters, and tire tracks are usually less than 0.5 meters. At this point, the terrain's original sample point can't depict the dynamic deformation of terrain effect. In other words, the sampling precision of the original terrain cannot meet the need of dynamic surface deformation. Therefore, higher precision is needed to realize the deformation effect of terrain. We call this dynamic terrain with multiscale, which is multiscale dynamic terrain. Aiming at the special situation of multiscale dynamic terrain, how to integrate high-precision deformation terrain with low-precision original terrain, data storage of high-precision deformation terrain and the influence of dynamic deformation on original low-precision terrain are the key issues for research.

Aiming at the special situation of multiscale dynamic terrain, this paper proposes a multiscale dynamic extension technique. Through the idea of multiscale dynamic extension, the high-precision deformation topography is extended to the original topography which needs to be processed, so that the terrain blocks with different precision are integrated. The deformation area data is stored by separate storage module. The triangular nodes that need to be drawn are selected by using nested sphere of error metric. And the constrained error radius gives the deformation area priority to draw.

II. TERRAIN VISUALIZATION ALGORITHM

In the rendering of 3d terrain visualization, the original terrain sampling data is stored by using triangle binary tree based on DAG[5, 6]. Using nested sphere of error metric to determine which node to draw per frame. And the algorithm of frame coherence and cone removal is also used. So that the requirements of real-time visualization of terrain rendering, and can better reflect the multiscale dynamic terrain.

A. The data structure: Triangular binary tree based on DAG

The selection of data structure is directly related to the size of terrain data storage and affects the speed of data access in real-time terrain rendering. Considering the advantages of binary tree in storage and access, we choose triangle binary tree to store the whole experimental terrain. The construction process is shown in figure 1.

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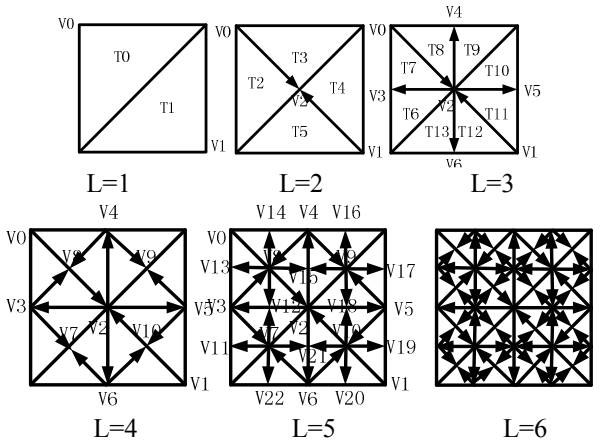


Fig. 1. Triangular binary tree process diagram

Figure 1 illustrates the splitting process of a six-layer binary tree. Division in the process, each connection right-angle vertex and the midpoint of the hypotenuse, father's triangle divided to two children. For example, T0 is divided into T2 and T3. And so on until the end of the whole terrain division, get the terrain of the triangular binary tree structure, as shown in figure 2.

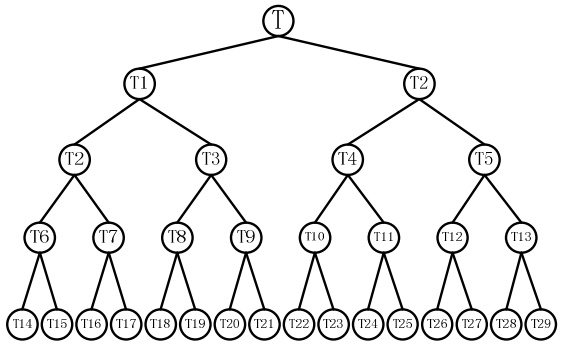


Fig. 2. binary tree storage

Figure 2 illustrates the storage structure of the four-layer binomial tree, continuing to split until the entire terrain binomial tree storage structure is established. Triangle binary tree is used in the algorithm to store the basic sampling data of the original terrain. In order to facilitate access of terrain data, and also to facilitate the use of nested sphere of error metric, a directed acyclic graph DAG is constructed according to the vertex introduction order in the generation of binary tree.

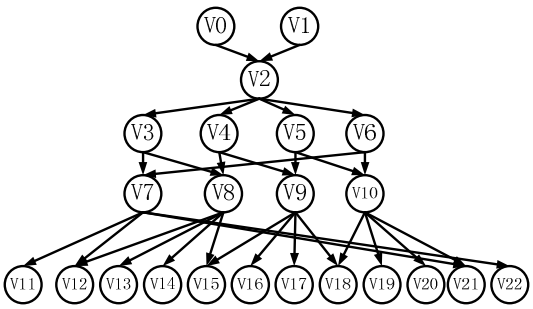


Fig. 3. GAD

To facilitate access and access of topographic data, and also to facilitate the use of nested error criterion balls, a directed noncyclic graph DAG with vertices is constructed according to the vertex introduction order in the generation of binary tree. And to use the chain DAG to store the vertices, so that the real-time mapping phase can quickly and efficiently access the terrain data.

In figure 3, Vi is the vertex, and the introduction order of the vertex constitutes the GAD graph. And you can see that each vertex has two parent vertices, four child nodes. And two father vertex each other brothers. To avoid cracks and t-connections, make sure the father's vertex is split when the child needs to split, that is, the two vertices are brothers. That is, the two vertices, which are brothers to each other, also divide at the same time. This makes the error evaluation function of construction will contain the relationship between implicit, thus effectively avoid the cracks and T - connection.

B. Error evaluation: nested sphere of error metric

The error evaluation function determines the number of triangles to be drawn for each frame. It not only affects the speed of real-time terrain rendering, but also affects the simulation degree of terrain rendering[6]. Therefore, the error evaluation function is very important. In this paper, the n nested sphere of error metric is used as the error evaluation function of real-time terrain rendering. It is mainly considered that when the child node needs to be split, the father node is guaranteed to be split, as shown in figure 4.

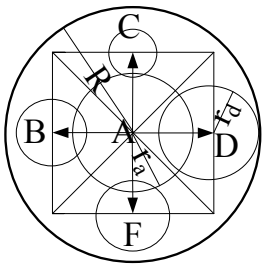


Fig. 4. nested sphere of error metric

As shown in figure 4, the father vertex A has four child vertices B, C, D and F. The original error of the father vertex is calculated to obtain the sphere of error metric with radius of r_a. That means the vertex A splits when the viewpoint enters the sphere of error metric with vertex A as the center of the circle and radius r_a. Similarly, taking child node D as an example, vertex D splits when the viewpoint enters the sphere with vertex D as the center of the circle and radius r_b. In order to avoid cracks and t-connection, it is required that when the child's vertices B, C, D and F are split, the father node A is split. Therefore, the nested sphere of error metric of the father vertex A is corrected, so that the error criterion ball of the father vertex contains the error criterion ball of all children's vertices, namely the nested sphere of error metric, the radius is R.

The calculation formula of the error criterion for the vertex related to the viewpoint is [7,8]:

$$r_i = \|p_i - e\| = \frac{\delta_i \times d \times \lambda}{\tau} \times k_i \qquad (1)$$

Among them, i is the vertex, λ represents the number of pixels corresponding to the unit length, τ represents the spatial error of the screen; P, e is the location of the nodes and points in the three-dimensional object space; d is the distance between the view point and the screen plane, δ indicates the object space error corresponding to the node. And $\delta_D=|Z_D-(Z_B+Z_C)/2|$.

As for the non-leaf vertex sphere of error metric, in addition to considering its own sphere of error metric, it also contains all children vertex sphere of error metric, so the nested sphere of error metric formula is defined as follows:

$$R_i = \begin{cases} r_i, & i = \text{leaf node} \\ \max(r_i, \max(\|p_i - p_j\| + R_j)), & \\ & i = \text{non} - \text{leafnode} \end{cases} \quad (2)$$

Among them, r_i is the radius of the error with roughness determined by formula 1, p_i, p_j is the space position of vertex I and j, vertex j is sub nodes of vertex i, C_i is the subset of the sub vertex of the vertex i in the DAG, Rj is the nested sphere of error metric for sub vertex j.

C. Algorithm to optimize

In the 3D terrain visualization algorithm, the real-time display of the terrain is very demanding. In the current large-scale topographic data, in order to achieve the purpose of fast rendering topography, some methods are usually used to optimize the algorithm, so as to improve the algorithm speed. In this paper, we use frame coherence and cone removal to optimize the visualization algorithm to improve the rendering efficiency.

Considering the high depth of binomial tree in topography, it takes time to construct binomial tree in every frame from the root node. Frame coherence, the main point of view is that when you're traveling in the terrain, you're not moving fast, and you don't have a lot of changes in the frame by frame. Thus, you can use the binary tree leaves in the first frame to do a proper merge or split properly to get the binary tree that needs to be rendered in this frame. In this way, the number of nodes of error judgment in each frame is reduced, and the algorithm efficiency is improved effectively[6].

The cone is removed, mainly considering that the observer can only see part of the terrain at some point. Therefore, the efficiency of topographic mapping algorithm can be effectively improved by eliminating the topographic node in the part beyond the visual cone [9,10]. Currently used visual cone consists of six surfaces.

The frame coherence and cone removal can effectively reduce the calculation amount of each frame, as well as the number of triangles to be drawn in each frame, so as to improve the algorithm efficiency and meet the requirements of real-time terrain rendering.

III. MULTISCALE DYNAMIC TERRAIN ALGORITHM

Aiming at the multiscale dynamic terrain, it is necessary to use higher precision to reflect the deformation effect of local topography, and at the same time, reduce the calculation amount after dynamic change of terrain to ensure the efficiency of real-time rendering[11]. Therefore, the main problem in the study of multiscale dynamic terrain is how to store multiscale deformation data after dynamic and how to integrate high-precision deformation areas into original low-precision terrain.

A. Generation of multiscale dynamic terrain

During the real-time rendering of terrain, there are ruts left by cars, craters left by bombs and footprints left after walking over soft ground. Therefore, the study of dynamic terrain with real-time changes of topographic surface elevation and texture color is especially meaningful.

The View-dependent LOD technology uses a low-resolution rendering of area surface far from the viewpoint, and high-resolution rendering of area surface close to the viewpoint. But the highest resolution is the same everywhere on the terrain, depending on the original minimum sampling interval on the terrain surface. For example, in the terrain provided in literature [12], the terrain sampling interval of crater is 1m/s, while that of island is 30m/s. Therefore, in the sampling of topographic data, the sampling interval is usually in meters.

However, the dynamic terrain changes, such as ruts and footprints, are generally much smaller than the terrain. It is difficult to characterize the deformation effect of the footprints if they are in the sampling topography of 1 meter, even the terrain with a 30-metre sampling interval is negligible. But for these dynamic deformation areas in the process of real-time rendering, it is precisely to depict and embody. Therefore, in order to really show the changes of dynamic terrain, more sampling points should be dynamically added between the original minimum sampling interval, that is, the number of local regional data sampling points should be increased. The resolution of deformation region and original terrain in the same terrain is inconsistent, as shown in figure 6. Since LOD technology is also known as multi-resolution technology, this situation of improving the multi-resolution of terrain caused by increasing regional sampling points becomes multiscale dynamic terrain.

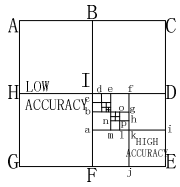


Fig. 5. multiscale dynamic terrain

As shown in figure 5, vertex ABCD and so on are the sampling points of the original terrain with low accuracy. In order to better reflect the deformation effect of terrain, the accuracy of terrain was improved in the original unit block at the lower right corner. The addition of vertices such as abcd allows for multiscale in a terrain. On the one hand, how to save the added vertex information efficiently. On the other hand, after improving the accuracy, new vertices, such as vertex abcd and so on, appeared in the deformation region and original terrain boundary, which would cause cracks and t-connection in the boundary. Therefore, in order to better represent the deformation effect of dynamic terrain, it is necessary to study and discuss the storage of multiscale dynamic terrain, error evaluation calculation, feature maintenance of deformation area and algorithm performance optimization. Among these problems, the impact on the original topography after the local region is improved, and how to avoid the crack and t-connection in the boundary region is the key problem to be studied.

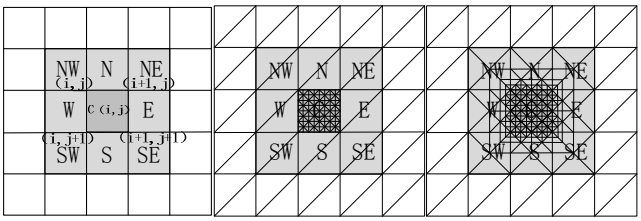
B. The problem of multiscale dynamic terrain

After the dynamic deformation of topography occurs, the attribute value of local terrain changes, which leads to multiscale dynamic terrain. In order to match the local high accuracy of terrain, the original terrain blocks near local

terrain also need to improve their accuracy. This will bring about changes in the extent of the original terrain, and how to avoid cracks and t-connection after changing in the boundary region, which is the key problem of multiscale dynamic terrain rendering.

For the convenience of description, the smallest square formed by four adjacent sampling points of the minimum sampling interval of the original topography is taken as the basic processing unit when the precision is extended, which is called the cell. It is also assumed that the extended resolution is always 2^k (k is a positive integer) times the original maximum resolution, and k is called precision index.

Definition 1: cell C (i, j) as the sample point (i, j), (i + 1, j), (i, j + 1), (i + 1, j + 1) of the square.



(a)cell C (b) deformation (c) transition zone

Fig. 6. multiscale dynamic deformation

As shown in figure 6(a), C(i, j) at the central location is a cell, and the cells adjacent to it are defined as W, N, E, S, NW, NE, SE, SW cells in turn. If cell C improves the accuracy to 23 = 8 times of the original cell, 7 points are added in the longitudinal and transverse directions of cell C, as shown in figure 6(b). Cells W, N, E, S that have common edges with cell C have added multiple original vertices of W, N, E, and S on the common edge (except the angular points of the cell). TTherefore, in order to avoid cracks and t-connection at the adjacent boundaries of cells, the accuracy of common edges should be kept consistent. And that means that the cell of W, N, E and S also need to be improved. In this way, does the accuracy of cells adjacent to them need to be adjusted after the improvement of cell W, N, E and S? The results show that the range of accuracy adjustment is not extended all the time. These cells that need to be adjusted are called cell transition zone, as shown in figure 6(c).

Definition 2: transition zone: the set of cells adjacent to a cell is called the transition region of the cell [3]. A cell where data changes is called a change cell. The cells in the transition area are called extension cells.

When the data precision in cell C is increased by 8 times, the precision of surrounding cells also needs to be expanded. In addition, only the transition zone precision composed of cells W, N, E,S, NW, NE, SE and SW around cell C needs to be extended. And it has no relation to the degree of data accuracy improvement in the cell C. That is, cells W, N, E,S, NW, NE, SE,SW are transition zones of cell C. When the accuracy of cell C changes, changes are needed and only cells W, N, E,S, NW, NE, SE,SW are needed.

Definition 3: dynamic region: the set of deformation cell C and its transition zone cells W, N, E,S, NW, NE, SE and SW are defined as dynamic region.

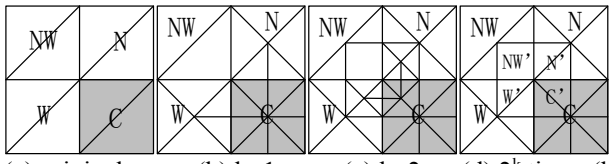
Theorem 1: suppose the precision of cell C is increased to 2^k times as before. In order to avoid cracks, no matter how large positive integer k is, only the resolution of cells in transition zone should be enlarged [3].

Therefore, after the accuracy of cell C is increased by 2^k times, in order to avoid cracks, only the precision of cells in transition zone should be expanded. As can be seen from the figure, the precision in the expanded area can also be determined.

Theorem 2: suppose that the precision of cell C is increased by 2^k times, and the mulscale of the precision of cell in transition zone is: the precision of cell with common edge is increased by 2^k times, and that of other cells is increased by 2^{k-1} times.

Prove: If the precision of cell C is increased by 2^k times, the precision of four edges is also increased by 2^k times. Since cell W, N, E, S and cell C have common edges, the accuracy of cell W, N, E and S can be increased to 2^k times as the precision of these edges can be increased to 2^k times. So, expand public edge cell accuracy for original 2^k times was established

The cells NW, NE, SE, SW at the other four corners, according to symmetry, have the same maximum precision. Take cell NW as an example, consider its extended precision multiscale. As shown in figure 7(a), when k=1, $2^{k-1} = 2^0 = 1$, that is, the accuracy of cell NW remains unchanged, and the conclusion is established. As shown in figure 7 (b), when k=2, $2^{k-1} = 2^1 = 2$, that is, the accuracy of cell NW is doubled, and the conclusion is established. Assuming that the conclusion is true when k=m (m>1), it is necessary to prove that k=m+1 is also true. As shown in figure 7 (d), when cell C expands by 2^{m+1} times, cell C can be divided into four. The precision per bisector is clearly2^m. Take the child cell in the upper left corner as C' for example. When m>1, the cells N, NW and W also have four child cells N', NW' and W'. When the child cell is C', the accuracy is obviously 2^m times, which is true according to the assumption that the precision of NW' is 2^{m-1} times. Since NW' is a quarter of NW, the precision of NW is 2^m times. From this, it can be concluded that when cell C can be expanded to 2^k times, the accuracy of cell NW is 2^m times, so k=m+1 is established.



(a) original (b) k=1 (c) k=2 (d) 2^k times (k≥2)

Fig. 7. precision extension diagram

It can be seen from the above research that when the accuracy of cell C is improved, only the cell precision of the transition zone should be improved accordingly, and other cells of the original terrain should not be affected. This result makes it easy and feasible to deal with terrain after multiscale dynamic terrain.

C. *Multiscale dynamic terrain processing*

It can be seen from the preceding analysis that for the multiscale dynamic terrain, dynamic cells and transition zones need to be processed and drawn. Therefore, when multiscale dynamic occurs in the real-time rendering stage of terrain, only the data of the dynamic region need to be processed. Triangular binary tree based on DAG will be used to store dynamic region data. The error evaluation function is used to simulate the deformation effect. The dynamic region is well integrated into the original terrain, so as to finish the

processing of multiscale dynamic terrain and achieve the purpose of terrain simulation drawing.

1) data structure

In order to facilitate the real-time rendering of dynamic region and the integration with the original terrain, the DAG based triangle binary tree is still used for data storage of dynamic region. Although it has been shown that cells NW, NE, SE and SW have a lower accuracy than cells in other deformable regions, they are still stored using full binary trees for convenience.

In order to make the cells in the transition region not affect the deformation effect of the dynamic cell, an approximate plane of the area can be provided in the dynamic region. In this way, the deformation effect of dynamic cells can be better expressed.

In the dynamic region, no elevation value can be provided in the transition region cell except for the dynamic cell C, which is provided with specific data due to deformation.

Definition 4: elevation value of dynamic region: for new vertices of no deformation area in transition region, the elevation value is uniformly defined as the average value of elevation value of all original terrain in dynamic region, as shown in formula 3.

$$Z_D = \frac{1}{16}\sum_{i=1}^{16} Z_i \qquad (3)$$

The dynamic region consists of 9 cells and 16 sampling points of the original terrain. Therefore, the elevation value of dynamic region is the average value of these 16 original terrain sampling points. Through uniform assignment, a nearly planar original region is formed inside the dynamic region except the original sample point.

2) Error calculation

During real-time terrain rendering, when multi-precision dynamic deformation occurs, all new vertices need to be calculated for error. In this paper we still use nested sphere of error metric to calculate the error of dynamic region. Moreover, the nesting thought can avoid the cracks and t-connection in the dynamic region

The state of the original sample point in the dynamic region is changed after the local shape has multiscale dynamic terrain. It may be that the original sampling points in the dynamic region do not need to be drawn, but after dynamic deformation, these sampling points need to be drawn. Therefore, in addition to recalculating the error of the sampling points in the dynamic region, the error of the ancestor vertex of the sampling points should be recalculated.

However, in order to reflect the deformation effect of dynamic region, the terrain in the dynamic cell should be emphatically displayed after the local shape has multiscale dynamic terrain. However, the original nested sphere of error metric only considers the viewpoint, so the deformation area may be weakened. In order to maintain the deformation effect of the dynamic region, the paper [2] sets the error of the vertices in the dynamic region uniformly as a maximum value, so that these vertices are forced to be displayed. This method can guarantee the effect of terrain deformation, but can not use LOD ideas internally, will produce too many triangles.

So, the constraint factors based on roughness are constructed by using the deformation effect of dynamic region. Therefore, the probability of dynamic region can be improved by using constraint factors, so that dynamic region can be highlighted first. Considering that the whole deformation process is real-time terrain rendering, all vertex storage and error calculation are completed in real-time rendering. Therefore, simple calculation is needed to complete the calculation of constraint factors to guarantee the efficiency of real-time rendering. As shown in formula 4, the ratio between the maximum elevation value difference in the dynamic cell and the maximum elevation value difference in the transition zone is used to reflect the degree to which the deformation region needs to be displayed in the original terrain. When the ratio is large, it indicates that the dynamic cell has obvious effect compared with the transition zone, which needs to be highlighted. Conversely, when the ratio is small, it indicates that the dynamic cells are not very effective compared with the transition zone. However, considering that the deformation effect of the surrounding terrain is not obvious, it still needs to reflect the deformation effect of the multiscale dynamic terrain. Therefore, when the ratio is less than 1, the error will be calculated according to the original nested sphere of error metric. In the formula 4, to ensure that the denominator is not zero, add ξ to the denominator and ξ is the minimum.

$$K_i = \max\left(\frac{\max(Z_a - Z_b)}{\max(Z_c - Z_d) + \xi}, 1\right) \begin{matrix} a,b \in cell(i) \\ c,d \in TZ(i) \end{matrix} \qquad (4)$$

Using the constraint factor constraint error criterion ball, it is redefined as formula 5.

$$R_i = \begin{cases} k_i r_i, & \text{i = leaf node} \\ \max(r_i, \max(\|p_i - p_j\| + k_i R_j)), & \\ & \text{i = non - leafnode} \end{cases} \qquad (5)$$

To sum up, when multiscale dynamic deformation occurs in real-time rendering, it is necessary to display dynamic cells emphatically. By constructing the error of the transition region and the extended deformation region, the main purpose of the dynamic region is shown. The error of the ancestor vertex of the original sample point should be recalculated to avoid the cracks and t-connection.

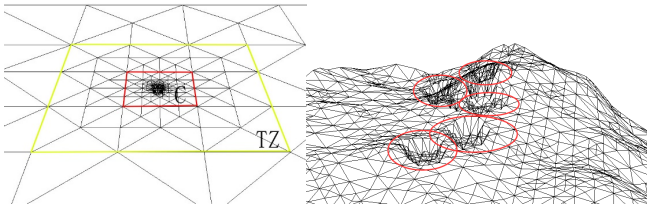
IV. Experimental Results

Algorithm about large-scale terrain based on this paper has been implemented on the computer. Hardware environment: Intel(R) Core(TM) i3-5-5u, 4G RAM, NVIDI2 A GeForce2400 graphics card; software environment: Windows 10, VS2010, and OpenGL.

A. Effect of deformation

In the experiment, the original plane terrain was used to dynamically generate the simulation effect of footprint deformation in real-time rendering of terrain. As shown in figure 11, the small (red) boxes in the figure are dynamic cells C, the transition zone between small boxes and large (yellow) boxes, and the dynamic region within the whole large boxes.

It can be seen from figure 8(a) that, in the real-time rendering process of terrain, when multiscale dynamic deformation of cell C occurs, the precision of cell C is improved, and the precision of surrounding cells is also improved, but the improvement of precision only extends to the transition zone. Therefore, only the terrain of the entire dynamic region can improve the precision of the display without affecting the original precision of other regions.

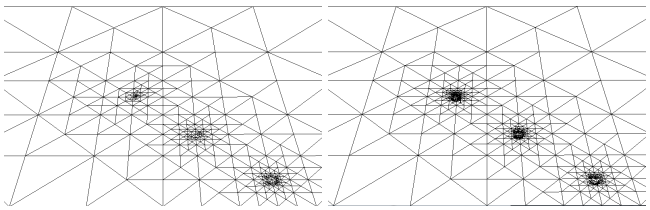


(a) deformation of footprint (b) dynamic deformation

Fig. 8. multiscale dynamic deformation

In figure 8(b), using the crater terrain [12], the original sample value is 513*513, showing the terrain local effect of the experimental terrain with multiscale dynamic terrain. It can be seen from the figure that the deformation effect is obvious, and the dynamic region can be well integrated into the original terrain.

Figure 9 shows the deformation effect comparison of the error calculation methods with and without constraints under the same viewpoint. It can be seen that constraint factor is used to constrain the nested sphere of error metric to achieve the purpose of preferential display of dynamic region.

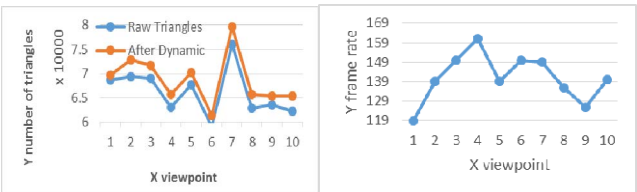


(a) error with constraints (b) error without constraints

Fig. 9. comparison of dynamic region under constraint factors

B. *The algorithm efficiency*

Since the multiscale dynamic deformation occurs in the real-time rendering of terrain, the creation of multiple data structures and the calculation of vertex error need to be completed in the real-time rendering stage. This will increase the workload of real-time rendering of the terrain and reduce the running efficiency of the algorithm.



(a) number of triangles increased (b) frame rate

Fig. 10. efficiency of multiscal dynamic deformation

However, due to the multiscale dynamic deformation of the terrain, the changes occur only in dynamic region. Therefore, the relative data amount is not large, and the error calculation is simple and convenient, which is suitable for real-time rendering of the terrain. The crater terrain[12] was used for experimental verification. Figure 10(a) shows the increased number of triangles when the terrain has multiscale dynamic deformation. Figure 10(b) shows the frame rate when the terrain has multiscale dynamic deformation.

It can be seen from figure 10(a) that the number of triangles generated by multiscal deformation is significantly less than the original number of large-scale terrain. And figure 10(b) shows that the frame rate can still meet the needs of real-time rendering when multi-precision dynamic deformation occurs.

CONCLUSION

Aiming at the situation of multiscale dynamic terrain, this paper eliminates the processing method and corresponding algorithm of multiscale dynamic terrain. After determining that the improved precision area is dynamic region, a triangle binary tree based on DAG is used to store the data of dynamic region, and the error radius is constrained, so that the multiscale dynamic terrain can be displayed preferentially. Experimental results show that this method can give priority to displaying multiscale dynamic region and can meet the requirements of real-time terrain rendering.

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